

Total No. of Questions : 8]

[Total No. of Printed Pages : 2

Roll No

EX-6003 (CBGS)

B.E. VI Semester

Examination, December 2020

Choice Based Grading System (CBGS)

Microprocessor and Microcontrollers

Time : Three Hours

Maximum Marks : 70

Note: i) Attempt any five questions.

ii) All questions carry equal marks.

1. a) Explain the architecture of 8086 microprocessor with a neat block diagram.
b) Explain the 8086 flag register with each flag bit.
2. Draw the timing diagram for read and write cycle in minimum mode 8086.
3. a) Draw the register organization of 8086 and explain application of each register.
b) Explain the following instruction of 8086 with example.
XCHG, DAA, XOR, PUSH, POP, RCR, AAA, INRET, OUT
4. a) What is the difference between 8254 and 8255 programmable interval timer? List the various modes of 8255. Explain any one mode with waveforms.
b) Draw the architecture of DMA controller. Explain in brief.
5. a) With the help of example explain programming of external interrupts.
b) Explain the memory structure of 8051.

EX-6003 (CBGS)

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[2]

6.
 - a) Draw and explain 8051 connections to RS-232.
 - b) Draw and explain the bit pattern of TMOD registers.
7.
 - a) With the help of example explain programming of external interrupts.
 - b) What do you mean by orthogonal instruction set?
8. Write a short notes on any two:
 - a) Programmable counter array (PCA)
 - b) Features and limitation of 32 bit PIC microcontroller
 - c) 8096 instruction set
